

smriti स्मृति

Memory that knows **when**.

The zero-infrastructure, bi-temporal memory layer for AI agents. One SQLite file — no Postgres, no Neo4j, no Docker, no cloud account.

one SQLite file

stdlib + numpy

Apache-2.0

MCP-ready

\$50K pre-seed

THE PROBLEM

Agents forget — and the usual fix is a data center.

To give one agent memory today

- ✗ Postgres + pgvector, or Neo4j for the graph
- ✗ Qdrant / Redis for vectors and cache
- ✗ Docker, a cluster to keep alive and version-match
- ✗ A cloud account — and graph/temporal behind a paywall

...and it **still** can't tell you what was true **last month**.

Delete-on-write stores overwrite the past. Accuracy silently decays as sessions accumulate.

Every memory framework makes you adopt infrastructure before you store a single fact.

THE PRODUCT

One SQLite file. Zero infrastructure.

The entire dependency surface is the Python standard library plus `numpy`. Around 2,000 readable lines, Apache-2.0, every feature included — no paid tier for graph, temporal, or scale.

1

file to back up

2

dependencies

0

servers to run

```
$ pip install -e .  
→ a working memory layer
```

HOW IT WORKS

Correction is an event, not an overwrite.

● ● ● ● memory.py

```
mem.add("I moved to Bengaluru on June 1st.")

mem.context("where do I live?")

# KNOWN FACTS:
# — [2026-01-15 | SUPERSEDED on 2026-06-01] The user lives in Hyderabad.
# — [2026-06-01 | CURRENT] The user lives in Bengaluru.
```

Facts are superseded, never deleted — the old fact is marked past and kept. One store answers "what's true now" and "what was true then."

ARCHITECTURE

Four retrieval rivers, one confluence.

shabda

शब्द · word

Lexical BM25 over FTS5

artha

अर्थ · meaning

Semantic vectors in numpy

sambandha

सम्बन्ध · relation

Entity-graph hop

kala

काल · time

Temporal proximity

sangama संगम · confluence → RRF fusion → validity-annotated, budget-capped context

Named profiles — facts · relations · timeline · deep · auto — bundle the channels into per-query policies.

WHY THE NAME

The vocabulary is load-bearing.

Indian epistemology worked out a precise technical language for memory two millennia before vector databases. Each borrowed term earned its place because the classical meaning is the engineering meaning.

anubhava → samskara → smriti

Experience leaves impressions; recollection arises from impressions. That is the write path — memory derived from experience, not the experience itself.

badha · sublation

A later cognition invalidates an earlier one without erasing that it occurred. That is supersession, precisely.

sangama · confluence

Four rivers meeting. Four retrieval channels, one fused ranking. The name is the architecture.

PERFORMANCE

Fast, cheap, and predictable at scale.

3.2_{ms}

warm query at 12.5k
memories

78_{ms}

warm query at 312k
memories

~50k/s

sustained ingest rows/sec

~\$2.50

frontier reader+judge over
500 questions

Retrieval packs a fixed ~1,600-token context per answer — per-query cost stays in fractions of a cent, on small, inexpensive models.

Honest envelope: **excellent** to tens of thousands, **acceptable** to low hundreds of thousands. Numpy quantization is roadmap #2.

THE FIELD

Everyone made a bet — each carries a database tax.

FRAMEWORK	THEIR STRENGTH	THE TAX IT CHARGES
mem0	Mature extraction, 90k+ devs	Graph behind a \$249/mo tier; self-host wants Docker + Postgres + Qdrant
Zep / Graphiti	Bi-temporal knowledge graph — the right model	Heavy graph-DB infra; advanced features cloud-only
Supermemory	Full-stack, strong leaderboard claims	A system you trust rather than read; forgetting is internal, unauditable
Smriti	Temporal model + entity graph + hybrid retrieval	One SQLite file. ~2k lines you read in a sitting. Supersession built in.

Our synthesis: keep the temporal model and the entity graph — drop the database tax.

THE MOAT

Small enough to read. Honest enough to verify.

Readable, forkable

~2k lines you audit in a sitting. No black box — the temporal model, entity graph, and profiles are all in the open.

Ship-and-verify

The benchmark harness ships with the library. Run it on **your** data, with **your** judge — one command.

Everything Apache-2.0

Trust earned through transparency, not a leaderboard we graded ourselves.

SELF-RUN A/B · FIXED JUDGE

Observation stack **+4.8 overall**, temporal-reasoning **+17.6**

Per-type router: multi-session **+10.3** (p=0.046) — no knowledge-update regression

SMRITI ENTERPRISE

Enterprise is where the revenue lives.

Replaceable modules over an untouched core — zero core edits, still stdlib-only. Built for regulated, on-prem, edge and air-gapped agent builders.

Tri-temporal as-of

World validity, system knowledge, storage lifecycle — three axes, never one column.

Evidence receipts

Canonical, integrity-checkable receipts for the exact context delivered — with HMAC checkpoints.

Exact lineage

Episode→fact→observation derivation edges. Erasure follows lineage, not heuristics.

Retention & legal holds

Hold-aware erasure and scheduled sweeps that report what was held.

Profiles that fail closed

local · team · regulated — egress denied by default, with a machine-readable data-flow map.

Verified knowledge packs

Signed, immutable, read-only snapshots — verify-before-open, RRF across stores.

BUSINESS MODEL

Open core drives adoption; enterprise drives revenue.

ADOPTION ENGINE

Open source • Apache-2.0

pip install, one file, MCP drop-in
Benchmark harness in the box
Every feature — graph, temporal, scale
Developer trust compounds into pull

REVENUE

Enterprise • per deployment

LTS releases + security backports
Signed releases + SBOM / provenance
Regulated deployment profiles
Migration, threat-model & evidence support, SLAs

Priced per deployment / site, not per seat — governance is becoming a buying requirement, and we sell the assurance around a free core.

WHERE WE ARE

What's shipped, and what's next.

✓ SHIPPED

Core 0.3.2 — 85 offline tests, no keys

Enterprise modules — 39 offline tests, tri-temporal kernel

MCP server — 6 typed tools, lite-by-default

Agile retrieval 0.2.0 + 0.3.x hardening

→ NEXT

Full LongMemEval / LoCoMo fixed-judge numbers

Numpy int8 / binary quantization (~10–30× headroom)

3–5 design partners (regulated / edge)

Platform adapters (Claude Code, Cursor, OpenClaw)

THE ASK

\$50K pre-seed

To build enterprise onboarding and support infrastructure — so we move fast on adoption while the open-source core compounds developer trust.

Design-partner onboarding

3–5 regulated / edge deployments

Deployment assurance

reference architectures, signed releases

Support + LTS pipeline

the revenue muscle for enterprise

Capital-efficient by construction — there is no infrastructure to run. That's the entire thesis.

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Memory that knows **when**.

One SQLite file. Zero infrastructure. Apache-2.0. The best no-dependency memory stack in the world — and we can prove it on **your** data.

github.com/vn-envy/Smriti

Prepared for Bolt • \$50K pre-seed